

Iterative solution of the Boltzmann Transport Equation for magneto-thermoelectric transport

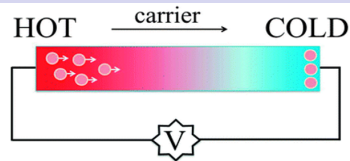
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What do we want ?



- We want to characterize charge and heat transfer in materials via the calculation of transport coefficients (σ , S , κ , ...).
- Fully first-principles approaches including electron-phonon coupling.
- BTE including temperature gradient is not that common.
- Magneto-thermoelectric transport ?

Characterize the thermoelectric transport

Electric current density:

$$\mathbf{j}^e = -\frac{\gamma e}{\Omega N_k} \sum_{nk} \mathbf{v}_{nk} f_{nk}$$

Heat current density:

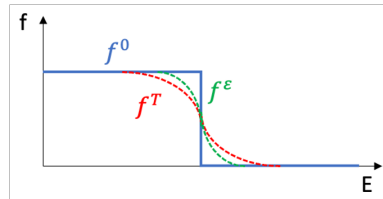
$$\mathbf{j}^Q = \frac{\gamma}{\Omega N_k} \sum_{nk} (\varepsilon_{nk} - \mu) \mathbf{v}_{nk} f_{nk}$$

Out of equilibrium distribution function:

$$f_{nk} = f^0(\varepsilon_{nk}, \mu, T) + \delta f_{nk}$$

where δf_{nk} is linear in the external fields and is a correction to Fermi-Dirac function:

$$\delta f_{nk} = \delta f_{nk}^{\mathcal{E}} + \delta f_{nk}^T$$



Boltzmann Transport Equation

The fundamental aim of the theory is the determination of the out-of-equilibrium statistical distribution function $f_n(\mathbf{r}, \mathbf{k})$.

Solving the BTE = determining $f \Rightarrow$ evaluation of transport coefficients.

The time-independent Boltzmann Transport Equation (steady state):

$$\underbrace{-\frac{\partial f_n}{\partial \mathbf{r}} \cdot \mathbf{v}_{nk}}_{\text{Diffusion}} - \underbrace{\frac{\partial f_n}{\partial \mathbf{k}} \cdot \mathcal{F}}_{\text{Forces}} + \underbrace{\mathcal{I}_{nk}[f]}_{\text{Collisions}} = 0$$

Phonon-Electron Scattering

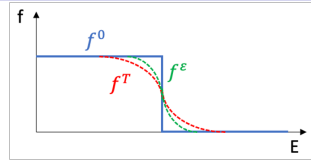


Linearized BTE (with $\mathbf{B} = 0$):

$$\frac{\partial f^0}{\partial \varepsilon_{nk}} \mathbf{v}_{nk} \left[-e\mathcal{E} - \frac{(\varepsilon_{nk} - \mu)}{T} \frac{\partial T}{\partial \mathbf{r}} \right] = \mathcal{I}_{nk}[f^0, \delta f]$$

Solving the BTE: IBTE method

- Iterative BTE (IBTE):



$$\frac{\partial f^0}{\partial \varepsilon_{nk}} \mathbf{v}_{nk} \left[-e\mathcal{E} - \frac{(\varepsilon_{nk} - \mu)}{T} \frac{\partial T}{\partial \mathbf{r}} \right] = -\frac{\delta f_{nk}}{\tau_{nk}^0} + \mathcal{L}_{nk}^{\text{e-ph}}[f^0, \delta f].$$

with τ^0 the relaxation time. f relaxes to equilibrium f^0 exponentially when fields are switched off, and

$$\delta f_{nk} = \delta f_{nk}^{\mathcal{E}} + \delta f_{nk}^T = \mathbf{F}_{nk}^{\mathcal{E}} \cdot \mathbf{E} + \mathbf{F}_{nk}^T \cdot \nabla T / T$$

$\mathbf{F}_{nk}^T$ and $\mathbf{F}_{nk}^{\mathcal{E}}$ are tensors giving the first derivatives of the distribution function f_{nk} with respect to ∇T and $\mathbf{E}$.

$\mathbf{F}$'s are calculated by solving the BTE at $\mathbf{E} = 0$ or $\nabla T = 0$ using an iterative scheme.

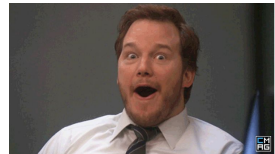
$$\begin{aligned}\mathbf{F}_{nk}^{\mathcal{E}(i+1)} &= e \frac{\partial f^0}{\partial \varepsilon_{nk}} \tau_{nk}^0 \mathbf{v}_{nk} + \frac{\tau_{nk}^0 2\pi}{N_q} \sum_{m\mathbf{q}\nu} |g_{mn\nu}(\mathbf{k}, \mathbf{q})|^2 \\ &\quad \times [(1 + n_{\mathbf{q}\nu} - f_{nk}^0)\delta^+ + (n_{\mathbf{q}\nu} + f_{nk}^0)\delta^-] \mathbf{F}_{m\mathbf{k}+\mathbf{q}}^{\mathcal{E}(i)} \\ \mathbf{F}_{nk}^T(i+1) &= \frac{\partial f^0}{\partial \varepsilon_{nk}} \tau_{nk}^0 \mathbf{v}_{nk} (\varepsilon_{nk} - \mu) + \frac{\tau_{nk}^0 2\pi}{N_q} \sum_{m\mathbf{q}\nu} |g_{mn\nu}(\mathbf{k}, \mathbf{q})|^2 \\ &\quad \times [(1 + n_{\mathbf{q}\nu} - f_{nk}^0)\delta^+ + (n_{\mathbf{q}\nu} + f_{nk}^0)\delta^-] \mathbf{F}_{m\mathbf{k}+\mathbf{q}}^T(i)\end{aligned}$$

Current densities:

$$\mathbf{j}^e = -\frac{\gamma e}{\Omega N_{\mathbf{k}}} \sum_{n\mathbf{k}} \delta f_{n\mathbf{k}} \mathbf{v}_{n\mathbf{k}} = -\frac{\gamma e}{\Omega N_{\mathbf{k}}} \sum_{n\mathbf{k}} \mathbf{v}_{n\mathbf{k}} \otimes (\mathbf{F}_{n\mathbf{k}}^{\mathcal{E}} \cdot \mathbf{E} + \mathbf{F}_{n\mathbf{k}}^T \cdot \nabla T / T)$$

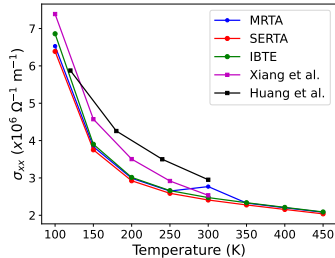
$$\mathbf{j}^Q = \frac{\gamma}{\Omega N_{\mathbf{k}}} \sum_{n\mathbf{k}} \delta f_{n\mathbf{k}} (\varepsilon_{n\mathbf{k}} - \mu) \mathbf{v}_{n\mathbf{k}} = \frac{\gamma}{\Omega N_{\mathbf{k}}} \sum_{n\mathbf{k}} (\varepsilon_{n\mathbf{k}} - \mu) \mathbf{v}_{n\mathbf{k}} \otimes (\mathbf{F}_{n\mathbf{k}}^{\mathcal{E}} \cdot \mathbf{E} + \mathbf{F}_{n\mathbf{k}}^T \cdot \nabla T / T)$$

⇒ One has the elements needed to calculate every thermoelectric transport coefficients in a "full" BTE way.



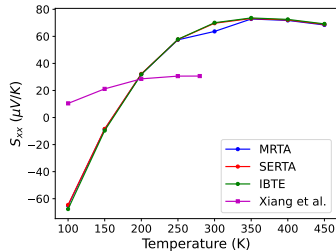
Results for Weyl semimetal TaAs

Electrical conductivity



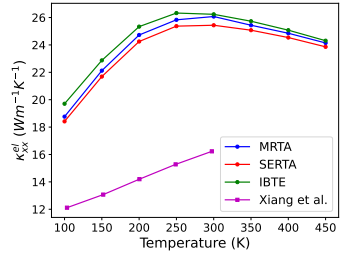
$$\sigma = \frac{j^e}{E}$$

Seebeck coefficient



$$S = \frac{E}{\nabla T}$$

Electronic thermal conductivity



$$\kappa^{el} = -\frac{j^Q}{\nabla T}$$

G. E. Allemand, M. Giantomassi, and M. J. Verstraete, Phys. Rev. B 112, 125122 (2025).

BTE with magnetic field

Same equation as before (with $\nabla T = 0$) + an additional term:

$$\underbrace{\left[1 - e\tau_{nk}^0(\mathbf{v}_{nk} \times \mathbf{B}) \cdot \frac{\partial}{\partial \mathbf{k}}\right]}_{\text{Magnetic term}} \delta f_{nk} = e\tau_{nk}^0 \frac{\partial f^0}{\partial \varepsilon_{nk}} \mathbf{v}_{nk} \cdot \mathcal{E} + \tau_{nk}^0 \mathcal{L}_{nk}^{\text{e-ph}}[f^0, \delta f]$$

Several difficulties:

- Calculate the $\mathbf{k}$ derivative (finite differences). ✓
- Found the k-points neighbors along each direction in a given energy range. ✓
- Symmetries breaking : TR and some spatial ones for anisotropic B fields !
→ $\mathbf{F}$ tensors asymmetric, what does it imply ? ✗

BTE with magnetic field, symmetries of conductivity tensor

Testing without symmetries (kptopt 3 and nsym 1) and without symmetrization routines:

- Diagonal symmetry under B: $\sigma_{xx}(\mathbf{B}) \approx \sigma_{yy}(\mathbf{B})$ ✓

- Hall antisymmetry: $\sigma_{xy}(\mathbf{B}) = -\sigma_{yx}(\mathbf{B})$ ✗

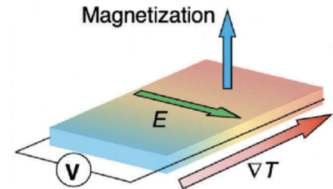
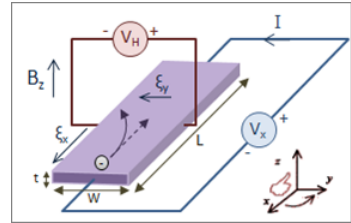
$$\sigma = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix}$$

- Onsager reciprocity: $\sigma_{xy}(\mathbf{B}) = \sigma_{yx}(-\mathbf{B})$ ✗

- Symmetry without B: $\sigma_{xy}(\mathbf{B} = 0) = \sigma_{yx}(\mathbf{B} = 0) = 0$ ✗ ✗ ✗

BTE with magnetic field ? What for ?

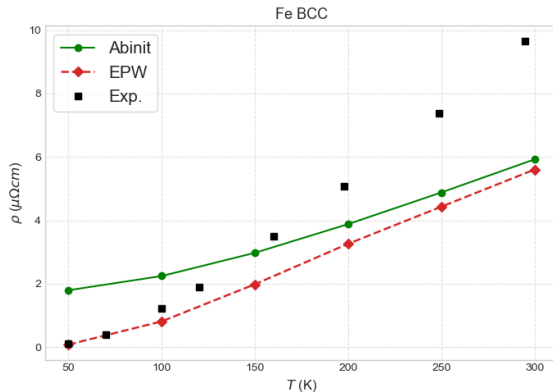
- Hall conductivity: BTE with $\mathbf{E}$ and $\mathbf{B}$
→ simply σ_{xy}
- Magnetoconductance: BTE with $\mathbf{E}$ and $\mathbf{B}$ along the same direction
→ simply σ_{xx} , σ_{yy} or σ_{zz}
- Nernst coefficient: BTE with $\mathbf{E}$, ∇T and $\mathbf{B}$
→ simply S_{yx}



We already have all the ingredients !

Transport calculations for magnetic materials

- EPH part for transport of Abinit does not work with nsppol 2
- Managed to enable the calculations
- Still to be checked, cleaned, and tested
- Incorrect behavior at low T



Á. A. Carrasco Álvarez et al., arXiv:2510.11350

Conclusion and perspectives

- Implementation of iterative BTE solution with E field and temperature gradient.
- Work still in progress for the BTE implementation with magnetic field, hope to be done with it soon (any symmetry and Abinit expert in the room ?)
- Investigate low T behavior for collinear spin calculations and merge the code.